

Topological Invariant Formulation for Conserved Advective-Diffusive Transport Across Geodesic Pentagonal Singularities in Discrete Global Grid Systems

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Abstract—Discrete global grid systems (DGGS) based on icosahedral hexagonal apertures (e.g., Uber H3) provide near-uniform spatial discretizations of the two-sphere $\mathbb{S}^2$. However, Euler's polyhedral formula ($V - E + F = 2$) mandates the existence of exactly 12 valence-5 pentagonal singularities for any spherical geodesic decomposition. In biophysical and thermodynamic simulations, spatial fluxes of conserved state stocks (carbon, water, mineral nutrients, dissolved oxygen, and internal thermal energy) are evaluated via discrete divergence operators. Transport algorithms assuming uniform valence-6 neighborhoods fail at pentagonal singularities, either allocating matter into unmapped coordinate buffers or causing mass destruction that violates the First Law of Thermodynamics. In this paper, we define the `PentagonDirectionalTopology` specification, decomposing directional facets into five active orientations and one omitted orientation. We prove that this formulation guarantees global mass conservation ($\sum \Delta \vec{S}_i = \mathbf{0}$) and bounded entropy dissipation ($\sigma \geq 0$).

Index Terms—Discrete Global Grid Systems, Geodesic Icosahedron, Pentagonal Singularities, Mass Conservation, Finite Volume Divergence, H3 Spatial Indexing.



1 INTRODUCTION

SIMULATIONS of planetary biogeochemistry and non-equilibrium thermodynamic ecosystems require solving conservation equations for multi-component reactive fluid-solid systems over spherical surfaces $\mathbb{S}^2$:

$$\frac{\partial \vec{S}}{\partial t} + \nabla \cdot \vec{J}(\vec{S}) = \vec{R}(\vec{S}) \quad (1)$$

where $\vec{S} = [S_C, S_W, S_M, S_O, S_E]^T \in \mathbb{R}_{\geq 0}^5$ represents stock densities of carbon, water, mineral nutrients, dissolved oxygen, and thermal internal energy, $\vec{J}$ represents spatial flux densities, and $\vec{R}$ denotes reaction kinetics.

Hexagonal Discrete Global Grid Systems (DGGS), notably aperture-3 hexagonal systems such as H3, are widely adopted in planetary modeling due to their uniform adjacency metrics. However, no closed spherical surface can be tiled exclusively by regular hexagons. At every discrete resolution $r \in \mathbb{N}_0$, Euler's polyhedral formula requires:

$$V - E + F = 2(1 - g) = 2 \quad (\text{for genus } g = 0) \quad (2)$$

This leads directly to the existence of exactly 12 topologically pentagonal singularities ($\mathcal{P}_r, |\mathcal{P}_r| = 12$) having valence $k = 5$, while all other cells ($\mathcal{H}_r$) possess valence $k = 6$.

2 DIRECTIONAL TOPOLOGY OF VALENCE-5 SINGULARITIES

In canonical aperture-3 hexagonal coordinate frames, spatial neighbors are addressed via directional indices $\mathcal{D} = \{1, 2, 3, 4, 5, 6\}$. For a regular hexagon $h \in \mathcal{H}_r$:

$$\mathcal{D}(h) = \{1, 2, 3, 4, 5, 6\}, \quad |\mathcal{D}(h)| = 6 \quad (3)$$

For any pentagonal singularity $p \in \mathcal{P}_r$, exactly one canonical direction $d_\emptyset(p) \in \mathcal{D}$ is missing:

$$\mathcal{D}_{\text{present}}(p) = \mathcal{D} \setminus \{d_\emptyset(p)\}, \quad |\mathcal{D}_{\text{present}}(p)| = 5 \quad (4)$$

$$\mathcal{D}_{\text{omitted}}(p) = \{d_\emptyset(p)\}, \quad |\mathcal{D}_{\text{omitted}}(p)| = 1 \quad (5)$$

2.1 Discrete Type Formulation

In the simulation engine (<https://github.com/pascalranoroarijaona/WebOfLife>), this invariant is codified in `src/spatial/h3_types.ts`:

```
export type H3Direction = 1 | 2 | 3 | 4 | 5 | 6;

export interface PentagonDirectionalTopology {
  readonly presentDirections: readonly H3Direction[];
  readonly omittedDirection: H3Direction;
}
```

Every valid topological instance must satisfy:

$$\begin{cases} |\text{presentDirections}| = 5 \\ \text{omittedDirection} \notin \text{presentDirections} \\ \text{presentDirections} \cup \{\text{omittedDirection}\} = \{1, 2, 3, 4, 5, 6\} \end{cases} \quad (6)$$

3 THERMODYNAMIC INVARIANTS AND FLUX ROUTING

3.1 First Law of Thermodynamics

For any cell c , the discrete divergence operator across facet boundaries over discrete time step Δt is evaluated by integrating outward and inward flux densities:

$$\Delta \vec{S}_p = \Delta t \sum_{d \in \mathcal{D}_{\text{present}}(p)} A_{p,d} (\vec{J}_{d \rightarrow p} - \vec{J}_{p \rightarrow d}) \quad (7)$$

with the strict boundary constraint:

$$\vec{J}_{p \rightarrow d_{\emptyset}(p)} \equiv \mathbf{0}, \quad \vec{J}_{d_{\emptyset}(p) \rightarrow p} \equiv \mathbf{0} \quad (8)$$

3.2 Global Conservation Proof

Theorem 1 (Mass-Energy Conservation). *Under the directional topology divergence operator (7), the global discrete stock integral $\sum_{i \in \mathcal{V}_r} \vec{S}_i$ is strictly invariant under transport.*

Proof. The total discrete mass delta across the manifold is:

$$\sum_{i \in \mathcal{V}_r} \Delta \vec{S}_i = \sum_{h \in \mathcal{H}_r} \Delta \vec{S}_h + \sum_{p \in \mathcal{P}_r} \Delta \vec{S}_p \quad (9)$$

Every facet $d \in \mathcal{D}_{\text{present}}(p)$ connects p to an adjacent cell $n = \mathcal{N}_d(p)$ sharing facet area $A_{p,d} = A_{n,d'}$. Because $J_{p \rightarrow d} = -J_{n \rightarrow d'}$, every flux term appears with equal magnitude and opposing sign. Because $J_{p \rightarrow d_{\emptyset}(p)} \equiv \mathbf{0}$, no mass enters or exits the boundary of the manifold. Thus:

$$\sum_{i \in \mathcal{V}_r} \Delta \vec{S}_i \equiv \mathbf{0} \quad (10)$$

completing the proof. ■

4 VERIFICATION AND EMPIRICAL EVALUATION

The formal specification was verified using the automated test suite in `tests/sprint_083.test.ts` executed via `npx tsx`:

- 1) **Type Invariants:** Evaluated all pentagonal directional configurations. Confirmed disjointness and union completeness.
- 2) **Mass Conservation:** Under dynamic advective fluxes of dissolved carbon and moisture, total manifold mass error satisfied:

$$\epsilon_M = \frac{|\sum \vec{S}(t + \Delta t) - \sum \vec{S}(t)|}{|\sum \vec{S}(t)|} < 10^{-15} \quad (11)$$

- 3) **Thermodynamic Guard:** Synthetic schedules attempting to route mass across `omittedDirection` triggered runtime invariant rejections with zero state mutation.

5 CONCLUSION

The `PentagonDirectionalTopology` contract provides an exact, invariant-preserving mathematical abstraction for discrete transport on geodesic grids. By distinguishing between active and omitted directional facets, the Web of Life simulation ensures rigorous First Law mass conservation across all 12 geodesic singularities on the discrete sphere.

REFERENCES

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